

11 (long)

A pipe in a power station heat exchanger carries gas at high pressure and temperature. The pipe is a thin-walled circular cylinder of diameter $D = 0.025$ m and wall thickness t (which is a design variable). The gauge pressure inside the pipe is p . The temperature of the gas inside the pipe is T . Outside the pipe, the gauge pressure is zero and the temperature is T_0 .

(a) A material is to be selected for the pipe from the options in Table 2. The rate of heat flow through the pipe wall per unit length ($\dot{Q}$) is to be maximised, where

$$\dot{Q} = (T - T_0) 2\pi\lambda \frac{D}{t}$$

and λ is the thermal conductivity of the pipe.

(i) If the hoop stress in the pipe σ_h must be less than the yield strength σ_y when $p = 3$ MPa, derive a material performance index that should be maximised, and hence identify the best material and the required wall thickness. [6]

(ii) A second constraint is added. A length $L = 2$ m of empty pipe is simply supported at both ends. It must deflect less than $\delta = 1.5$ mm mid-span due to self weight. Identify whether this constraint alters the material choice and the rate of heat transfer. [8]

(iii) Describe two other constraints to consider in the material selection, and comment on how they may influence the design and performance of the pipe. [4]

(b) A thin protective coating (thickness much less than t) is perfectly bonded to the inside of the pipe. The coating and pipe are strain free when $p = 0$ and $T = T_0$. When $p > 0$ and $T > T_0$, the pipe is supported such that the longitudinal stresses in the pipe wall remain zero, and there is no constraint against longitudinal expansion. The pipe has Young's modulus E_p and Poisson's ratio ν_p . For the coating, these are E_c and ν_c . The coefficients of thermal expansion of the pipe and coating are α_p and α_c , respectively.

(i) Derive an expression for the hoop strain ε_{hc} and longitudinal strain ε_{lc} in the coating when $p > 0$ and the coating and pipe wall are at a uniform temperature $T = T_0$. [4]

(ii) Derive an expression for the hoop stress σ_{hc} in the coating when $p > 0$ and the coating and pipe wall are at a uniform temperature $T > T_0$. [8]

	Cu alloy	stainless steel	Al alloy
Young's modulus, E (GPa)	130	200	75
Density, ρ (kg m^{-3})	8200	7850	2700
Yield strength, σ_y (MPa)	230	550	300
Thermal conductivity, λ ($\text{W m}^{-1} \text{K}^{-1}$)	280	20	160

Table 2